CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
First Semester of B.Sc.(Physics) Examination March 2018
PD 101 Mathematical Physics-I

Date: 26/03/2018, Monday **Time**: 10:00 a.m. to 10:30 a.m. **Maximum Marks**: 20

MCQ

Important Instructions:

- Provide all necessary information as required in the answer sheet.
- Attempt all questions.
- Answer the questions in the question paper itself.
- Use of cell phones is strictly prohibited
- Use of non-programmable calculators may be allowed if needed.

Q.1. Choose the correct answer for the following questions. [20]

1. What is the value of integral $\iint\limits_{S} dxdy$, where S is circle $x^2 + y^2 = 1$?

   a. $\frac{\pi}{2}$ b. π c. 2π d. 4π

2. Which of the following result established relation between surface integral and volume integral?

   a. Green's theorem c. Gauss's divergence theorem
   b. Stoke's theorem d. none of these

3. For a vector $\bar{a} = -3\hat{i} + 7\hat{j} + 5\hat{k}, \bar{b} = -3\hat{i} + 7\hat{j} - 3\hat{k}$ and $\bar{c} = 7\hat{i} - 5\hat{j} - 3\hat{k}$, $\bar{a} \cdot (\bar{b} \times \bar{c})=$ __________.

   a. 272 b. -272 c. 256 d. 234

4. Let $s(x, y, z) = c$ represent a surface. Then __________ represent normal to surface at point (x,y,z).

   a. $(\nabla \cdot s)_{(x,y,z)}$
   b. $(\nabla \times s)_{(x,y,z)}$
   c. $(\nabla s)_{(x,y,z)}$
   d. $(\nabla^2 s)_{(x,y,z)}$

5. A force field $\bar{F}$ is conservative, if __________.

   a. $\nabla \bar{F} = 0$
   b. $\nabla \cdot \bar{F} = 0$
   c. $\nabla \times \bar{F} = 0$
   d. $\nabla^2 \bar{F} = 0$

6. For a vector function $\bar{F}$, if $\nabla \cdot \bar{F} = 0$, then $\bar{F}$ is called __________.

   a. conservative
   b. solenoidal
   c. irrotational
   d. none of these

7. For a spherical coordinates (r, θ, ϕ), which of the following is true?

   a. $r < 0, \theta \geq 2\pi, \phi \geq 2\pi$
   b. $r \geq 0, \theta \geq 2\pi, \phi \geq 2\pi$
   c. $r \geq 0, \theta \geq 2\pi, \phi \geq \pi$
   d. $r \geq 0, \theta \geq \pi, \phi \geq 2\pi$

8. Which of the following probability distribution is continuous probability distribution?

   a. Normal distribution
   b. Binomial distribution
   c. Poisson distribution
   d. none of these

9. The variance for binomial distribution is__________.

   a. np b. npq c. $\sqrt{npq}$ d. nq,

   where n is repeated trails, p the probability of success and q is the probability of failure.

10. An urn contains four red balls, three green balls and five white balls. If a single ball is drawn, what is the probability that is green?

    a. 0.8 b. 0.75 c. 0.5 d. 0.25

11. If $P(A) = \frac{1}{3}, P(B') = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{6}$, then $P(A \cup B)$=__________.

    a. $\frac{11}{12}$ b. $\frac{1}{12}$ c. $\frac{5}{12}$ d. $\frac{7}{12}$

12. For normal distribution, $P(-\infty \leq z \leq \infty)$=__________.

a. 1 b. 0.5 c. 0.638 d. 0.754

13. Which of the following function is not differentiable at origin?

a. $\sin x$ b. $\frac{1}{x+1}$ c. e^{-x} d. $|x|$

14. If $\delta(t-a)$ is the Dirac delta function, then $\int_0^\infty \delta(t-3)(t^2+1)\,dt$=__________.

a. -8 b. -10 c. 10 d. 0

15. The order and degree of differential equation $(\frac{dy^5}{dx^5})^2 - xy = \cot x$ are

a. order=5, degree=2
b. order=2,degree=5
c. order=1,degree=5
d. order=5,degree=1

16. The differential equation satisfy by $y = A\sin t + B\cos t$ is __________, where A and B are arbitrary constants.

a. $\frac{d^2y}{dt^2} - y = 0$
b. $\frac{d^2y}{dt^2} + y = 0$
c. $\frac{d^2y}{dt^2} - y = \sin t$
d. $\frac{d^2y}{dt^2} - y = \cos t$

17. How many arbitrary constants are there in general solution of differential equation $y'' - 5y' + 6y = 0$?

a. 3 b. 2 c. 1 d. 0

18. $\lim_{x\to 2} \frac{x^2-4}{x-2}$=__________.

a. 0 b. 2 c. 4 d. 8

19. A real function f is said to be continuous at x_0, if__________.

a. $\lim_{x\to x_0} f(x) = 0$
b. $\lim_{x\to x_0} f(x) = f(x_0)$
c. $\lim_{x\to x_0} f(x) = x_0$
d. none of these

20. The integrating factor of linear differential equation $\frac{dy}{dx} + 2xy = \sin x$ is __________.

a. e^{x^2} b. e^{2x} c. e^x d. $e^{\frac{x}{2}}$

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

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PD 101 Mathematical Physics-I

Date: 26/03/2018, Monday **Time**: 10:30 a.m. to 1:00 p.m. **Maximum Marks**: 50

Instructions:

1. **Section I and II must be attempted in seperate answer sheets.**
2. **Make suitable assumptions and draw neat figures wherever required.**
3. **Use of non-programmable calculator is allowed.**
4. **Show necessary calculations.**
5. **Figures to right indicate marks.**

SECTION-I

Q-2 Answer the following questions as directed. [20]

(a) Find the volume of the parellopiped with adjacent sides, $\overline{OA} = 3\hat{\imath} - \hat{\jmath}$, $\overline{OB} = \hat{\jmath} + 2\hat{k}$ and $\overline{OC} = \hat{\imath} + 5\hat{\jmath} + 4\hat{k}$ extending from the origin of co-ordinates O. [02]

(b) Evaluate $\lim\limits_{n\to\infty} \frac{n^4-n^2+2n-9}{n^5-n^4+3}$. [03]

(c) Define Dirac delta function $\delta(x-5)$. Show that the area cover by curve of Dirac delta function and x-axis is one. [03]

(d) Find the Macluarin series expansion of $\cos 2x$. [03]

OR

(d) Using binomial expansion, find the approximate value of $(2 + 0.005)^3$. [03]

(e) Write cylindrical and spherical coordinates. [02]

(f) Evaluate the integral $\int\limits_0^1 \int\limits_0^x (x^2 + y^2) dA$, where dA indicates small area in xy-plane. [03]

OR

(f) Find the work done by a force $y\hat{\imath} + x\hat{\jmath}$ which displace a particle from origin to a point $(\hat{\imath} + \hat{\jmath})$ along a line $y = x$. [03]

(g) A random variable X is normally distributed with mean $\mu = 2$ and variance $\sigma^2 = 4$. Evaluate $P(1 < X < 3)$, if $P(0 < \frac{X-\mu}{\sigma} < 0.5) = 0.1915$. [02]

OR

(g) For a Poisson variate X, if $P(X = 3) = P(X = 4)$, then find $P(X = 0)$. [02]

(h) Solve the differential equation $\frac{d^2y}{dt^2} - 5\frac{dy}{dt} - 14y = 0$. [02]

OR

(h) Show that a differential equation $(x^2 - y^2)\,dx - 2yx\,dy = 0$ is exact. [02]

SECTION II

Q-3 Answer the following questions as directed. [30]

(a) Find the maximum value of $V(x, y, z) = xyz$ subject to the constraint $2x + 2y + z = 108$. [05]

(b) Two unbiased dice are tossed simultaneously. Find the probability that sum of numbers on the upper face of dice is 9 or 12. [05]

(c) Solve the differential equation $\frac{dy}{dt} - \frac{y}{t} = t^3 - 5t - 7$. [05]

OR

(c) Using method of variation of parameters, solve the differential equation $\frac{d^2y}{dt^2} - 7\frac{dy}{dt} + 12y = e^{5t}$. [05]

(d) Express $z\hat{i} - 2x\hat{j} + y\hat{k}$ in cylindrical coordinates. [05]

OR

(d) Prove that a spherical coordinate system is orthogonal. [05]

(e) Find the directional derivative of the function $\psi = x^2y^4 + z^2y^4 + x^2z^4$ at the point (2,0,3) in the direction of the outward normal to the sphere $x^2 + y^2 + z^2 = 14$ at the point (3,2,1). [05]

OR

(e) Define Solenoidal and irrotational vector function. Show that $(y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational. [05]

(f) Using Green's theorem, evaluate $\int_C (x^2y\,dx + x^2\,dy)$, where C is the boundary described counter clockwise of the triangle with vertices $(0,0), (1,0)$ and $(1,1)$. [05]

OR

(f) Evaluate $\int_C \bar{F} \cdot d\bar{r}$, where $\bar{F} = x^2\hat{\imath} + xy\hat{\jmath}$ and C is the boundary of the square in the plane $z = 0$ and bounded by the lines $x = 0, y = 0, x = 3$ and $y = 3$. [05]

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